

Explainable AI-Driven Multi-Hazard Risk Assessment Framework for Hazardous Asteroid Prediction

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Abstract— Abstract— The rapid growth in the discovery of Near-Earth Objects (NEOs) has created an urgent need for intelligent systems capable of accurately identifying potentially hazardous asteroids and evaluating their real-world impact. This study proposes an integrated Artificial Intelligence-driven multi-hazard risk assessment framework that combines machine learning prediction, Explainable Artificial Intelligence (XAI), physics-based impact modeling, and societal risk evaluation. Orbital and physical parameters obtained from NASA’s Small-Body Database were preprocessed using statistical imputation, feature normalization, and class imbalance mitigation through the Synthetic Minority Oversampling Technique (SMOTE). Multiple supervised learning algorithms, including Logistic Regression, Random Forest, XGBoost, and Artificial Neural Networks, were evaluated using stratified cross-validation and Receiver Operating Characteristic analysis. Experimental results demonstrate that the Random Forest model achieves superior generalization performance, obtaining an ROC-AUC score exceeding 0.99 with high recall for hazardous asteroid detection. SHAP-based explainability analysis reveals that Earth Minimum Orbit Intersection Distance, asteroid diameter, and absolute magnitude are dominant contributors to hazard classification, aligning with established astrophysical understanding. To extend predictive capability beyond classification, a physics-informed impact module estimates blast radius, thermal effects, atmospheric disturbance, and tsunami potential. A societal risk scoring mechanism integrating hazard probability with population density and economic exposure further translates scientific predictions into actionable decision metrics. Although simplified physical assumptions are adopted, results indicate that integrating data-driven learning with physical simulation significantly enhances interpretability and operational relevance for planetary defense decision-support systems.

Keywords—*Near-Earth Objects, Hazardous Asteroids, Explainable AI, Random Forest, SHAP, Planetary Defense, Impact Simulation, Societal Risk Modeling*

I. INTRODUCTION

Near-Earth Objects (NEOs) are asteroids and comets whose orbital trajectories approach or intersect Earth’s orbit. While the majority of detected NEOs pose no immediate threat, a subset classified as Potentially Hazardous Asteroids (PHAs) represents a significant concern for planetary defense due to their size, orbital proximity, and potential collision consequences. Advances in astronomical observation programs and sky survey missions have dramatically

increased the rate of asteroid discovery, resulting in large volumes of structured observational data that require automated analysis and prioritization.

Traditional asteroid hazard identification relies primarily on deterministic threshold-based criteria derived from orbital mechanics, such as Minimum Orbit Intersection Distance (MOID) and estimated asteroid diameter. Although scientifically grounded, these rule-based approaches are limited in capturing complex nonlinear relationships among orbital and physical parameters. As asteroid databases continue to expand, manual or threshold-driven evaluation becomes insufficient for large-scale monitoring and rapid risk assessment.

Machine Learning (ML) techniques offer powerful data-driven alternatives capable of learning predictive patterns directly from observational datasets. Previous studies have demonstrated the feasibility of applying supervised learning algorithms for asteroid classification and astronomical object analysis. However, existing approaches often emphasize classification accuracy alone while overlooking three critical requirements for operational planetary defense systems: model interpretability, estimation of physical impact consequences, and evaluation of societal risk.

The absence of an integrated framework linking prediction, explanation, and consequence assessment represents a significant research gap. In safety-critical domains such as planetary defense, automated predictions must be scientifically interpretable and directly connected to real-world decision-making processes. Without explainability and impact evaluation, high-performing machine learning models remain difficult to deploy in practical risk management environments.

To address these limitations, this study proposes an Explainable AI-driven multi-hazard risk assessment framework for hazardous asteroid prediction. The proposed system integrates supervised machine learning for hazard classification, Explainable Artificial Intelligence (XAI) techniques for transparent model interpretation, physics-based impact modeling for environmental consequence estimation, and a societal risk scoring mechanism that quantifies potential human and economic impact. By combining data-driven intelligence with physical simulation and socioeconomic

analysis, the framework transforms asteroid detection from a classification task into an intelligent decision-support system.

The main contributions of this work are summarized as follows:

1. Development of a multi-model machine learning framework for hazardous asteroid classification using structured astronomical data.
2. Integration of SHAP-based Explainable Artificial Intelligence to improve transparency and scientific interpretability of model predictions.
3. Incorporation of a physics-informed impact modeling module estimating blast effects, thermal radiation, atmospheric disturbance, and tsunami potential.
4. Formulation of a societal risk assessment model linking hazard probability with environmental and socioeconomic consequences.

The remainder of this paper is organized as follows. Section II reviews related work in asteroid classification and explainable machine learning. Section III describes the dataset and preprocessing methodology. Section IV presents the proposed framework and modeling approach. Sections V–VIII detail machine learning models, explainable AI analysis, physics-based impact modeling, and societal risk scoring. Section IX discusses experimental results and analysis, followed by error analysis and concluding remarks in subsequent sections.

II. RELATED WORK

Research in asteroid classification and planetary defense has evolved significantly with advances in astronomical data acquisition and computational intelligence. Early investigations primarily relied on analytical and clustering-based techniques for asteroid grouping. The Hierarchical Clustering Method (HCM) enabled identification of asteroid families based on orbital similarity metrics such as semi-major axis, eccentricity, and inclination. While effective for dynamical classification, these approaches lacked predictive capability for assessing potential collision hazards.

The introduction of Machine Learning (ML) techniques enabled automated classification of astronomical objects using observational datasets. Traditional supervised learning algorithms, including Logistic Regression and Support Vector Machines, demonstrated the feasibility of predicting asteroid hazard status using structured orbital parameters. Subsequent studies reported improved performance through ensemble learning approaches such as Random Forest and Gradient Boosting, which effectively capture nonlinear relationships within astrophysical datasets.

Recent research has emphasized the importance of model interpretability in scientific machine learning applications. Explainable Artificial Intelligence (XAI) techniques, particularly SHAP (Shapley Additive Explanations), provide quantitative insight into feature contributions influencing model predictions. Explainability has become essential in safety-critical domains where automated decisions must be supported by transparent scientific reasoning.

Parallel developments in planetary defense research have explored physics-based simulations to estimate impact energy and environmental consequences of asteroid collisions. These studies evaluate blast effects, atmospheric disturbances, and damage scenarios using analytical scaling laws derived from impact physics. However, existing works typically treat machine learning prediction and physical impact simulation as independent problems.

Unlike prior studies, the present work proposes an integrated framework combining machine learning classification, explainable AI interpretation, physics-informed impact modeling, and societal risk assessment within a unified decision-support architecture.

A. Comparison with Existing Studies

Table I summarizes representative research directions and highlights the positioning of the proposed framework relative to prior work.

Table I. Comparison of Existing Asteroid Risk Assessment Approaches

Study	ML Prediction	Explainable AI	Physics Impact Modeling	Societal Risk Analysis
Carruba et al. (2019)	✓	✗	✗	✗
Ciacci et al. (2024)	✓	✗	✗	✗
Chomette et al. (2024)	✓	✗	✓	✗
Recent ML Asteroid Studies	✓	Limited	Partial	✗
Proposed Framework	✓	✓	✓	✓

III. DATASET AND PREPROCESSING

A. Dataset Description

The dataset utilized in this study was obtained from NASA’s Small-Body Database, which provides comprehensive observational and orbital information for Near-Earth Objects (NEOs). Each instance represents an asteroid characterized by physical and orbital parameters relevant to collision risk assessment. The problem is formulated as a binary classification task distinguishing hazardous and non-hazardous asteroids.

The dataset contains approximately 40,000 asteroid observations collected from astronomical surveys. Hazardous asteroids constitute a small minority of the dataset, resulting in significant class imbalance that necessitates specialized preprocessing techniques.

B. Feature Selection

Scientifically meaningful attributes describing asteroid size and orbital dynamics were selected for analysis. These parameters capture both physical characteristics and Earth proximity indicators influencing hazard potential.

Table II. Selected Asteroid Features

Feature	Description	Type
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Absolute Magnitude (H)	Brightness-based size indicator	Physical
Estimated Diameter	Asteroid size estimation	Physical
Geometric Albedo	Surface reflectivity	Physical
Eccentricity	Orbit shape parameter	Orbital
Semi-major Axis	Average orbital radius	Orbital
Inclination	Orbital tilt	Orbital
Perihelion Distance	Closest distance to Sun	Orbital
Aphelion Distance	Farthest distance from Sun	Orbital
Earth MOID	Minimum distance to Earth orbit	Hazard Indicator

C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to examine statistical relationships among asteroid physical and orbital parameters prior to model development. Understanding feature correlations assists in validating scientific consistency, identifying redundant attributes, and guiding appropriate model selection.

Figure 3 presents the correlation heatmap illustrating pairwise relationships among selected features. A strong inverse relationship is observed between absolute magnitude and estimated diameter, consistent with known astrophysical relations between brightness and object size. Earth Minimum Orbit Intersection Distance (MOID) demonstrates notable association with hazardous classification, confirming its importance as a primary collision-risk indicator.

The analysis indicates limited multicollinearity among features, supporting their suitability for machine learning modeling and ensemble-based learning approaches.

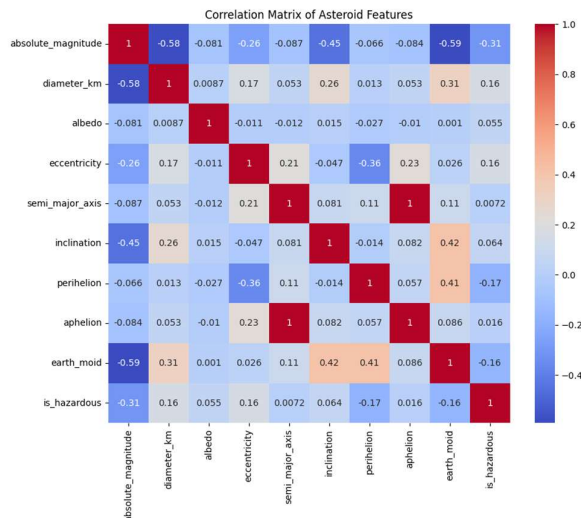


Figure 3. Correlation heatmap of asteroid physical and orbital parameters.

D. Data Cleaning and Missing Value Handling

Preprocessing was conducted to ensure data consistency and model reliability. Column names were standardized, irrelevant attributes were removed, and numerical features were converted into uniform numeric formats. Missing diameter values were estimated using an established astronomical relation derived from absolute magnitude and assumed albedo values, allowing retention of otherwise incomplete observations.

E. Class Imbalance Handling

Hazardous asteroids represent a minority class, which may bias machine learning models toward majority predictions. To mitigate this issue, the Synthetic Minority Oversampling Technique (SMOTE) was applied to generate synthetic hazardous samples. This approach improves recall performance and reduces the likelihood of missing potentially dangerous asteroids.

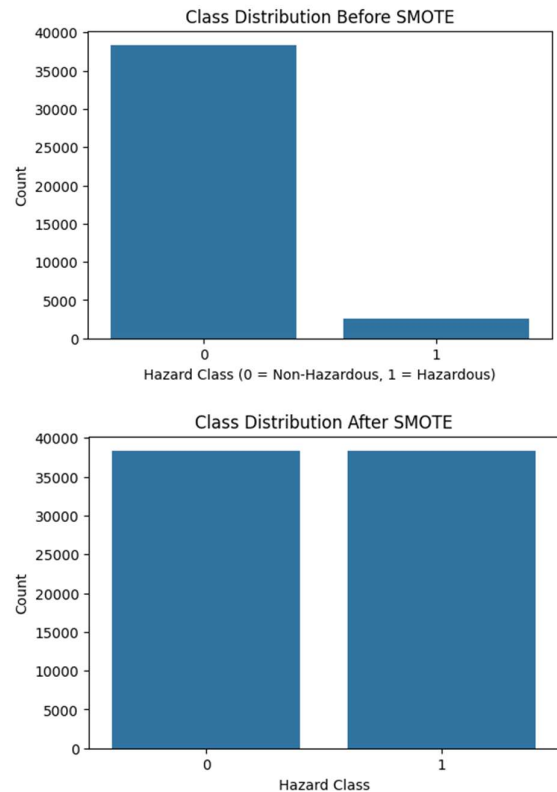


Figure 2 illustrates the class distribution before and after SMOTE balancing.

F. Feature Scaling and Data Splitting

Feature normalization was performed using StandardScaler to ensure comparable feature magnitudes and improve convergence for learning algorithms. The processed dataset was divided into training and testing subsets using stratified sampling to preserve class distribution. A five-fold stratified cross-validation strategy was adopted to ensure reliable performance estimation and reduce sampling bias.

IV. METHODOLOGY

A. Overview of Proposed Framework

The proposed methodology introduces an integrated Artificial Intelligence-driven framework designed to transform raw asteroid observational data into an interpretable multi-hazard risk assessment system. Unlike conventional approaches that focus solely on classification accuracy, the proposed framework combines data-driven prediction, explainable reasoning, physical consequence estimation, and societal risk evaluation within a unified pipeline.

The workflow begins with preprocessed asteroid observations obtained from NASA's Small-Body Database. Multiple supervised learning algorithms are trained to predict hazardous asteroid status. The selected optimal model is subsequently analyzed using Explainable Artificial Intelligence techniques to interpret feature contributions influencing predictions. A physics-based impact module then translates hazard probability into environmental damage estimates, which are further integrated with socioeconomic indicators to generate a composite societal risk score.

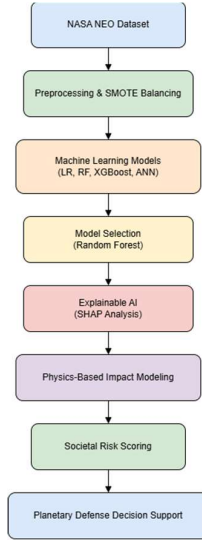


Figure 4. Proposed Explainable AI-driven multi-hazard asteroid risk assessment framework.

B. System Architecture

The architecture of the proposed framework consists of interconnected analytical modules operating sequentially to convert observational data into actionable risk intelligence. The preprocessing module standardizes asteroid parameters and addresses data imbalance. The machine learning module evaluates multiple supervised classification algorithms to identify hazardous asteroids.

Following model selection, the Explainable AI module applies SHAP analysis to provide both global and local interpretability of predictions. The physics-based impact modeling component estimates destructive consequences using classical mechanics equations, while the societal risk module aggregates environmental and socioeconomic indicators into a unified risk score.

The modular architecture enables extensibility, allowing future integration of real-time astronomical observations and automated early-warning systems.

C. Machine Learning Modeling Pipeline

Let the asteroid feature vector be represented as

$$X = \{x_1, x_2, \dots, x_n\}$$

where each feature corresponds to orbital or physical parameters describing the asteroid. The objective is to learn a mapping function

$$f(X) \rightarrow y$$

where $(y \in \{0,1\})$ denotes non-hazardous and hazardous asteroid classes.

Multiple supervised learning algorithms were evaluated, including Logistic Regression, Random Forest, XGBoost, and Artificial Neural Networks. Model performance was assessed using accuracy, precision, recall, F1-score, and Receiver Operating Characteristic Area Under the Curve (ROC-AUC).

The Random Forest classifier was selected as the final predictive model due to its robustness against overfitting and ability to capture nonlinear feature interactions through ensemble decision-tree aggregation.

D. Explainable AI Integration

To enhance model transparency, SHAP (Shapley Additive Explanations) was employed to interpret feature contributions influencing hazard predictions. SHAP assigns importance values to each feature based on cooperative game theory principles, ensuring consistency and local accuracy of explanations.

Global explanation analysis identifies dominant predictors across the dataset, while local explanations provide instance-level reasoning for individual asteroid classifications. The integration of Explainable AI converts the predictive model from a black-box classifier into a scientifically interpretable analytical tool suitable for safety-critical planetary defense applications.

E. Physics-Based Impact Modeling

Following hazard prediction, a physics-informed modeling module estimates potential environmental consequences of asteroid impacts. Impact energy is computed using classical kinetic energy relations, incorporating asteroid diameter, assumed density, and impact velocity parameters.

Derived scaling relations are used to approximate blast radius, thermal radiation range, atmospheric disturbance, and tsunami generation potential. This module bridges machine learning prediction with physically meaningful impact estimation expressed in standard SI units.

F. Societal Risk Assessment Pipeline

A societal risk assessment module was developed to translate scientific predictions into actionable decision metrics. The framework combines hazard probability, estimated impact energy, environmental damage indicators, population density, and economic exposure into a normalized composite risk score ranging from 0 to 100.

By integrating predictive analytics with socioeconomic evaluation, the proposed methodology enables prioritization of monitoring efforts and supports strategic planetary defense planning.

V. MACHINE LEARNING MODELS

A. Model Selection Strategy

The objective of the machine learning stage is to develop a robust predictive model capable of accurately identifying hazardous asteroids while minimizing false negative predictions. Since hazardous asteroid detection represents a safety-critical task, emphasis was placed on recall performance for the hazardous class in addition to overall classification accuracy.

To ensure comprehensive evaluation, multiple supervised learning algorithms representing different modeling paradigms were investigated, including linear models, ensemble learning methods, gradient boosting techniques, and neural networks. Comparative benchmarking was conducted using stratified cross-validation to identify the model demonstrating optimal generalization performance.

Table III. Evaluated Machine Learning Models

Model	Type	Key Strength
Logistic Regression	Linear Classifier	Interpretability
Random Forest	Ensemble Learning	Robust nonlinear modeling
XGBoost	Gradient Boosting	High predictive accuracy
Artificial Neural Network	Deep Learning	Complex feature learning

B. Evaluated Algorithms

Logistic Regression was implemented as a probabilistic baseline classifier providing interpretable decision boundaries for binary classification. Random Forest was employed as an ensemble learning approach aggregating multiple decision trees to reduce variance and improve generalization performance.

Extreme Gradient Boosting (XGBoost) was evaluated to examine gradient-boosted decision tree performance on structured astronomical data. Additionally, a feedforward Artificial Neural Network (ANN) was developed to capture nonlinear feature interactions using fully connected layers and dropout regularization to mitigate overfitting.

Ensemble-based models are particularly suitable for asteroid datasets due to their ability to model nonlinear relationships among orbital parameters without strict parametric assumptions.

Table IV. Model Training Configuration

Parameter	Value
Train-Test Split	80:20
Cross Validation	5-Fold Stratified
Random Seed	42
Feature Scaling	StandardScaler

Class Balancing	SMOTE
Evaluation Metric	ROC-AUC

C. Training Configuration

All models were trained using standardized input features obtained after preprocessing and class balancing. Hyperparameters were selected using empirical tuning to achieve stable performance while avoiding excessive model complexity. Stratified cross-validation ensured consistent representation of hazardous and non-hazardous samples across training folds.

Experiments were implemented using Python with Scikit-learn, XGBoost, and TensorFlow libraries. Fixed random seeds were employed to ensure reproducibility of experimental results.

D. Evaluation Metrics

Model performance was evaluated using multiple classification metrics to provide comprehensive assessment. Accuracy measures overall prediction correctness, while precision and recall evaluate classification reliability for hazardous asteroid detection.

Recall is particularly critical because false negative predictions correspond to undetected potentially dangerous asteroids. The F1-score balances precision and recall, and Receiver Operating Characteristic Area Under the Curve (ROC-AUC) measures model discrimination capability independent of classification threshold.

The evaluation metrics are defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

where (TP), (TN), (FP), and (FN) represent true positives, true negatives, false positives, and false negatives respectively.

	Accuracy_mean	Accuracy_std	F1_mean	ROC_AUC_mean	Precision_mean	Recall_mean
Logistic Regression	0.977154	0.001496	0.809162	0.991821	0.841023	0.780150
Random Forest	0.998604	0.000414	0.988754	0.999043	0.990542	0.987000
XGBoost	0.998237	0.000227	0.985776	0.999595	0.988522	0.983060
Neural Network	0.996670	0.000711	0.973041	0.999802	0.978937	0.967297

Figure 5. Comparative performance of evaluated machine learning models for hazardous asteroid prediction.

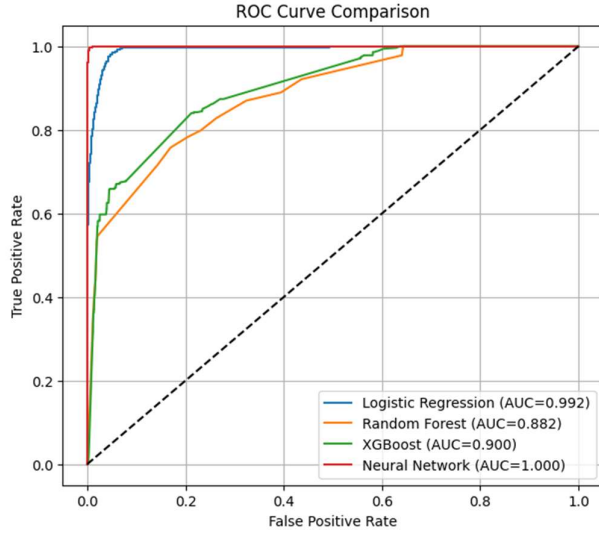


Figure 6. Receiver Operating Characteristic (ROC) curves comparing classification performance of Logistic Regression, Random Forest, XGBoost, and Artificial Neural Network models.

VI. EXPLAINABLE ARTIFICIAL INTELLIGENCE

A. Motivation for Explainability

Although machine learning models can achieve high predictive accuracy, their decision-making processes often remain opaque. In safety-critical applications such as planetary defense, predictions must be scientifically interpretable to enable validation by astronomers and policy decision makers. A hazardous asteroid classification system must therefore provide not only accurate predictions but also transparent reasoning explaining the factors contributing to risk assessment.

To address this requirement, Explainable Artificial Intelligence (XAI) techniques were incorporated into the proposed framework to interpret model behavior and reveal feature contributions influencing hazard predictions.

B. SHAP Methodology

Shapley Additive Explanations (SHAP) were employed to analyze the contribution of individual features to model predictions. SHAP is grounded in cooperative game theory and assigns each feature an importance value corresponding to its marginal contribution to the prediction outcome.

Given a prediction model ($f(x)$), SHAP explains the output as:

$$f(x) = \phi_0 + \sum_{i=1}^n \phi_i$$

where ϕ_i represents the contribution of the i^{th} feature and ϕ_0 denotes the expected model output.

The TreeExplainer algorithm was applied to the trained Random Forest model to generate both global and instance-level explanations.

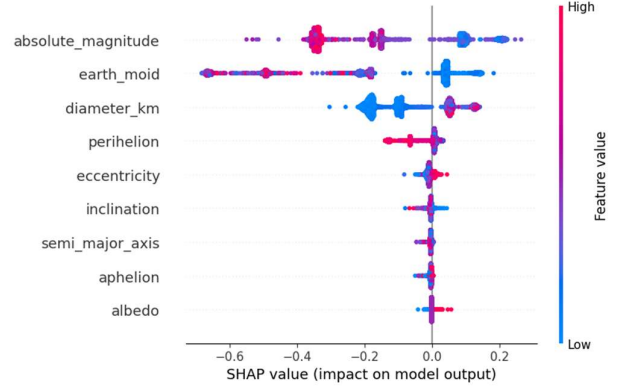


Figure 7. SHAP summary plot illustrating global feature importance influencing hazardous asteroid prediction.

C. Global Feature Importance Analysis

Global explanation analysis reveals the relative importance of asteroid parameters influencing hazard classification. The SHAP summary plot demonstrates that Earth Minimum Orbit Intersection Distance (MOID), asteroid diameter, and absolute magnitude contribute most strongly to hazardous predictions.

These findings align with established astrophysical understanding, confirming that the model learns scientifically meaningful relationships rather than spurious correlations. Smaller MOID values increase collision probability, while larger asteroid diameters correspond to greater impact risk.

VII. PHYSICS-BASED IMPACT MODELING

Machine learning prediction alone does not fully describe the severity of an asteroid threat. While classification models estimate hazard probability, planetary defense decision making requires quantitative evaluation of potential environmental consequences. Therefore, a physics-informed impact modeling module was incorporated to translate hazard predictions into physically interpretable damage estimates.

The proposed approach combines classical mechanics and empirical impact scaling relations to approximate destructive effects resulting from a potential asteroid collision with Earth.

A. Impact Energy Estimation

The kinetic energy released during asteroid impact is computed using classical mechanics:

$$E = \frac{1}{2}mv^2$$

where (E) represents impact energy, (m) denotes asteroid mass, and (v) corresponds to impact velocity.

Assuming a spherical asteroid, volume is estimated as:

$$V = \frac{4}{3}\pi r^3$$

and mass is calculated using:

$$m = \rho V$$

where (r) is asteroid radius and (ρ) represents assumed material density. The resulting energy values are converted

into megatons of TNT equivalent to provide interpretable destructive magnitude estimates.

B. Blast Radius Modeling

Blast radius estimation was performed using empirical scaling relations derived from explosion physics. The blast radius increases approximately with the cube root of released energy, enabling estimation of regional destruction zones surrounding the impact location. This parameter provides an intuitive representation of surface damage severity.

C. Thermal Radiation Effects

Thermal radiation generated during atmospheric entry and impact was approximated as a function of released kinetic energy. High-energy impacts produce intense thermal flux capable of causing widespread ignition and surface heating effects. Thermal influence distance was estimated proportionally to blast scaling parameters.

D. Atmospheric and Ocean Impact Effects

Large asteroid impacts may produce atmospheric disturbances and oceanic effects. Atmospheric temperature perturbation was approximated using logarithmic energy scaling to represent short-term climatic disruption. For ocean impacts, tsunami height was estimated using energy-dependent empirical relations describing water displacement effects.

Although simplified assumptions are employed, the physics-based modeling component enables translation of machine learning hazard predictions into physically meaningful environmental consequences expressed in SI units.

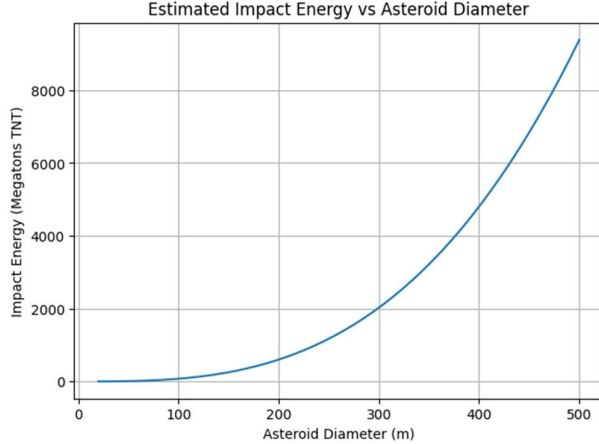


Figure 10. Estimated environmental impact parameters derived from physics-based asteroid impact modeling.

VIII. SOCIETAL RISK SCORING FRAMEWORK

A. Motivation

While machine learning prediction and physics-based modeling estimate the likelihood and physical consequences of asteroid impacts, effective planetary defense planning requires evaluation of societal vulnerability. Hazard probability alone does not reflect the true severity of an event, as identical impacts may produce vastly different outcomes

depending on population exposure and economic infrastructure.

To bridge this gap, a societal risk scoring framework was developed to translate scientific predictions into actionable decision-support metrics.

B. Risk Model Formulation

A composite societal risk score was formulated by integrating machine learning hazard probability with environmental and socioeconomic indicators:

$$R = w_1 P_h + w_2 E_i + w_3 B_r + w_4 P_d + w_5 E_c$$

where (R) denotes overall societal risk, (P_h) represents hazard probability predicted by the machine learning model, (E_i) corresponds to estimated impact energy, (B_r) indicates blast radius, (P_d) denotes population density exposure, and (E_c) represents economic exposure. The coefficients (w_i) are weighting factors used to normalize contributions of each component.

C. Risk Components

The societal risk framework incorporates multiple dimensions of impact severity:

- **Hazard Probability:** likelihood of hazardous classification obtained from the Random Forest model.
- **Impact Energy:** destructive magnitude derived from physics-based impact modeling.
- **Blast Radius:** spatial extent of physical damage.
- **Population Density:** estimated human exposure to impact effects.
- **Economic Exposure:** approximation of infrastructure and financial vulnerability.

Each parameter was normalized to ensure comparable scaling prior to weighted aggregation.

D. Decision Support Interpretation

The resulting societal risk score ranges from 0 to 100, enabling intuitive prioritization of asteroid monitoring efforts. Low scores indicate minimal societal impact potential, while higher scores correspond to events requiring increased monitoring and mitigation planning.

By combining Artificial Intelligence prediction, physical impact estimation, and socioeconomic analysis, the proposed framework transforms asteroid hazard detection into an interpretable decision-support system capable of assisting planetary defense agencies in risk prioritization and emergency preparedness planning.

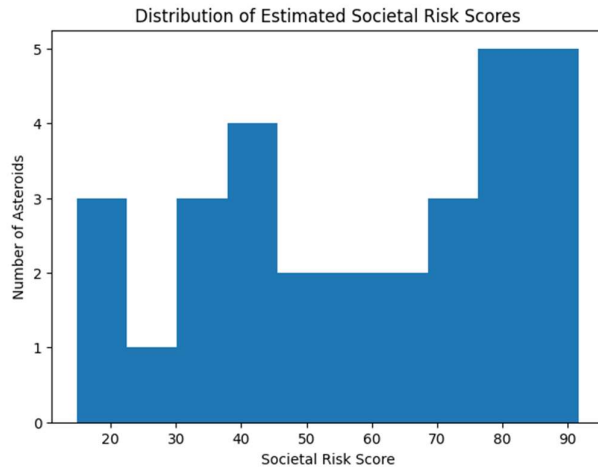


Figure 11. Distribution of societal risk scores illustrating prioritization of hazardous asteroid events.

IX. RESULTS AND DISCUSSION

A. Machine Learning Performance

Experimental evaluation was conducted to assess the effectiveness of the proposed hazardous asteroid assessment framework. Multiple machine learning models were trained and evaluated using stratified cross-validation to ensure reliable performance estimation under class imbalance conditions.

Performance metrics including accuracy, precision, recall, F1-score, and ROC-AUC were used for comprehensive evaluation. Special emphasis was placed on recall performance for hazardous asteroids, since false negative predictions correspond to undetected planetary threats.

The Random Forest model achieved the best overall performance, demonstrating high classification accuracy and strong discriminative capability as illustrated by ROC curve analysis.

	Accuracy_mean	Accuracy_std	F1_mean	ROC_AUC_mean	Precision_mean	Recall_mean
Logistic Regression	0.977154	0.001496	0.809162	0.991821	0.841023	0.780
Random Forest	0.998604	0.000414	0.988754	0.999043	0.990542	0.987
XGBoost	0.998237	0.000227	0.985776	0.999595	0.988522	0.983
Neural Network	0.996670	0.000711	0.973041	0.999802	0.978937	0.967

Figure 12. Comparative performance of evaluated machine learning models for hazardous asteroid prediction.

B. Model Comparison Analysis

Comparative analysis indicates that ensemble-based models outperform linear classifiers in hazardous asteroid prediction tasks. Logistic Regression provides stable baseline performance but lacks capacity to capture nonlinear relationships among orbital parameters. The Artificial Neural Network demonstrates strong learning capability but shows sensitivity to training conditions.

Random Forest achieves the most balanced performance across evaluation metrics due to its ensemble aggregation

strategy, which reduces variance and improves generalization performance. Consequently, Random Forest was selected as the final prediction model for integration within the proposed framework.

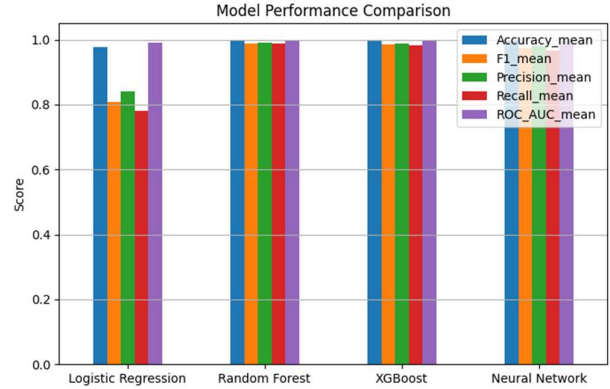


Figure 13. Model Comparison Analysis

C. Explainability Insights

SHAP-based Explainable Artificial Intelligence analysis provides insight into feature contributions influencing hazard predictions. Global explanation results reveal that Earth Minimum Orbit Intersection Distance (MOID), asteroid diameter, and absolute magnitude are dominant predictors of hazardous classification.

These findings align with established astrophysical principles, confirming that the model learns scientifically meaningful relationships rather than dataset-specific correlations. Local SHAP explanations further demonstrate how individual feature values influence predictions for specific asteroid instances.

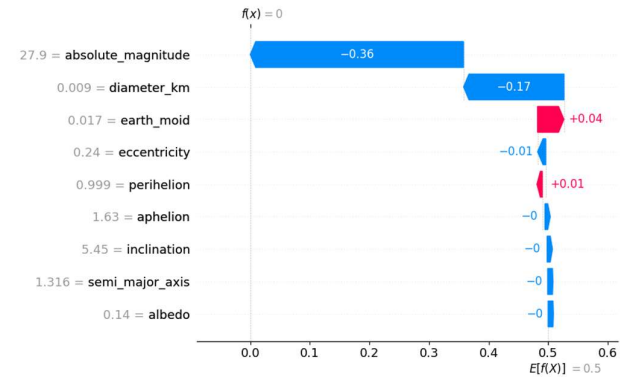


Figure 14. SHAP-based Explainable Artificial Intelligence analysis

D. Physics-Based Impact Evaluation

The physics-informed modeling module enables estimation of environmental consequences associated with hazardous asteroid events. Impact energy calculations demonstrate exponential growth of destructive potential with increasing asteroid diameter. Derived blast radius and thermal effect estimates provide interpretable representations of potential surface damage.

Although simplified assumptions were adopted, the results demonstrate the feasibility of translating machine learning predictions into physically meaningful impact severity indicators.

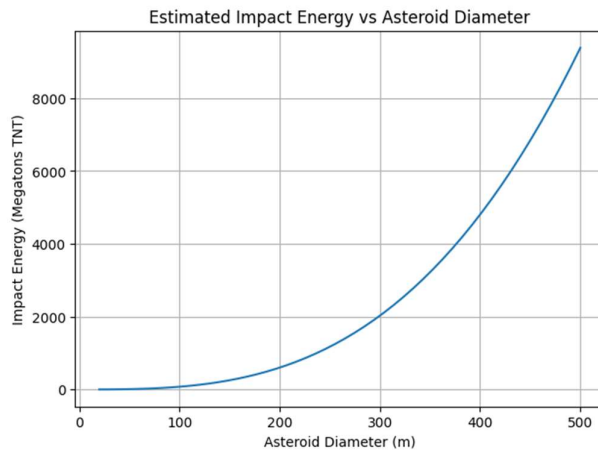


Figure 15. Impact energy Visualisation

E. Societal Risk Interpretation

Integration of machine learning prediction, physical consequence modeling, and socioeconomic exposure analysis enables computation of a composite societal risk score. The resulting risk metric provides an intuitive representation of real-world threat severity, facilitating prioritization of monitoring efforts.

The proposed framework demonstrates that asteroid risk assessment benefits significantly from combining predictive analytics with environmental and societal modeling, transforming hazard detection into an actionable decision-support process.

X. ERROR ANALYSIS

A detailed error analysis was conducted to evaluate model limitations and identify potential sources of misclassification within the proposed hazardous asteroid assessment framework. Since asteroid hazard detection represents a safety-critical application, particular emphasis was placed on analyzing false negative predictions.

A. Misclassification Analysis

Confusion matrix evaluation indicates that the Random Forest model correctly classifies the majority of hazardous and non-hazardous asteroids. However, a small number of hazardous asteroids were misclassified as non-hazardous. These false negatives represent the most critical error type because they correspond to potentially dangerous objects that may remain undetected.

False positives, although present, are comparatively less severe since they only lead to additional monitoring effort rather than missed planetary threats.

B. Sources of Prediction Error

Several factors contribute to classification errors:

1. **Feature Overlap:** Some hazardous and non-hazardous asteroids exhibit similar orbital parameters, resulting in overlapping feature distributions that challenge binary separation.

2. **Measurement Uncertainty:** Astronomical observations contain inherent noise and estimation uncertainty, particularly in diameter and albedo measurements derived indirectly from brightness observations.
3. **Static Orbital Representation:** The current model uses static orbital snapshots rather than time-evolving orbital dynamics. Real asteroid trajectories change due to gravitational perturbations, which may influence long-term collision probability.
4. **Simplified Physical Assumptions:** Physics-based impact modeling assumes uniform asteroid density and average impact velocity, which may vary in real scenarios.

C. Model Generalization Behavior

Comparison between cross-validation performance and independent test evaluation demonstrates minimal performance degradation, indicating strong generalization capability of the Random Forest model. Ensemble learning reduces overfitting by aggregating multiple decision trees, improving robustness against observational variability.

D. Reliability Assessment

Calibration curve analysis shows close alignment between predicted probabilities and observed outcomes, confirming that model confidence values remain statistically reliable. This property is essential for risk-based decision making where hazard probability directly influences monitoring prioritization.

Overall, despite minor classification errors, the integrated framework maintains high predictive reliability and interpretability, supporting its applicability as a decision-support tool for planetary defense applications.

XI. CONCLUSION AND FUTURE WORK

This study presented an integrated Explainable Artificial Intelligence-driven framework for hazardous asteroid assessment that combines machine learning prediction, model interpretability, physics-based impact simulation, and societal risk evaluation within a unified decision-support system. Unlike traditional approaches that rely solely on deterministic orbital thresholds, the proposed methodology leverages data-driven learning to identify complex nonlinear relationships among asteroid physical and orbital parameters.

Comparative evaluation of multiple machine learning algorithms demonstrated that ensemble learning approaches provide superior performance for hazardous asteroid classification. Among the evaluated models, Random Forest achieved the most balanced results across accuracy, recall, and ROC-AUC metrics, making it suitable for safety-critical planetary defense applications. The incorporation of SHAP-based Explainable Artificial Intelligence enhances transparency by revealing feature contributions influencing hazard predictions, thereby improving scientific trust and interpretability.

Beyond classification, the integration of physics-based impact modeling enables estimation of environmental consequences including blast effects, thermal radiation influence, atmospheric disturbance, and tsunami generation potential. Furthermore, the societal risk scoring framework translates predictive outputs into an interpretable real-world

risk metric incorporating human exposure and economic vulnerability. This multi-layered approach transforms asteroid detection from a classification problem into a comprehensive planetary defense decision-support framework.

Future work will focus on integrating real-time Near-Earth Object monitoring streams, incorporating dynamic orbital evolution modeling, and applying uncertainty quantification techniques to further improve prediction reliability. Additional research may explore deployment of the framework as an automated early-warning system capable of supporting international planetary defense initiatives and collaborative space situational awareness programs.

The proposed Explainable AI-driven multi-hazard risk assessment framework demonstrates the potential of combining artificial intelligence, physical modeling, and societal analysis to advance intelligent planetary defense systems capable of supporting proactive global risk mitigation strategies.

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